

8. Substituting in the equation $D^2U - \text{II} \cdot DU = 0$, we obtain as the conditions of a five-pointic contact

$$\begin{aligned} & - (m-2) \partial_1^2 U + P \cdot \frac{1}{2} \partial_1^2 U = 0, \\ & - \frac{1}{3} [(m-1) \partial_1^3 + 3(m-2) \partial_1 \partial_2] U + P \cdot \frac{1}{6} (\partial_1^3 + 3 \partial_1 \partial_2) U + \partial_1 P \cdot \frac{1}{2} \partial_1^2 U = 0, \\ & - \frac{1}{12} [(m-1)(\partial_1^4 + 6 \partial_1^2 \partial_2) + (m-2)(4 \partial_1 \partial_3 + 3 \partial_2^2)] U \\ & + P \cdot \frac{1}{24} [\partial_1^4 + 6 \partial_1^2 \partial_2 + 4 \partial_1 \partial_3 + 3 \partial_2^2] U + \partial_1 P \cdot \frac{1}{6} (\partial_1^3 + 3 \partial_1 \partial_2) U + \frac{1}{2} \partial_2 P \cdot \frac{1}{2} \partial_1^2 U = 0; \end{aligned}$$

or reducing

$$\begin{aligned} P &= 2(m-2), \\ \partial_1 P &= \frac{2}{3} \frac{\partial_1^3 U}{\partial_1^2 U}, \\ \partial_2 P &= \frac{1}{2} \frac{[\partial_1^4 + 6 \partial_1^2 \partial_2] U}{\partial_1^2 U} - \frac{4}{3} \frac{\partial_1^3 U}{\partial_1^2 U} \cdot \frac{[\partial_1^3 + 3 \partial_1 \partial_2] U}{\partial_1^2 U}, \end{aligned}$$

which are the conditions of a five-pointic contact: it is to be remarked that if only the first and second conditions are satisfied, we have a four-pointic contact, and if only the first condition is satisfied, a three-pointic contact.